

Three-Frame Multi-Line Alignment of Bovine GPRC6A Exon 4

This report wraps long alignments across multiple rows (showing 38 codons per row) to ensure full text readability. Codons and their amino acids remain stacked vertically. The target proteotypic barcode ("NDVFIVTNQETK") is highlighted in bold red.

Bovine Reading Frame +1 — [FRAME-SHIFTED / MUTATED]

Human (DNA/AA):	TTA L	CTT L	GGT G	GTG V	---	---	CTA L	AAA K	AAT N	GTG V	ACA T	TTC F	ACT T	GAT D	GGA G	TGG W	AAT N	TCA S	TTT F	CAT H	TTT F	GAT D	GCT A	CAC H	GGG G	GAT D	TTA L	AAT N	ACT T	GGA G	TAT Y	GAT D	GTT V	GTG V	CTC L	TGG W	AAG K	GAG E	
Match Alignment:																																							
Bovine (DNA/AA):	AGT S	ACT T	GGA G	GTG V	GGT G	TGC C	CAT H	TGC C	CTT L	CTC L	CGA R	TTC F	ATG M	GAT D	GGA G	GGG G	AGT S	TCA S	TTT F	CAT H	TTT F	TAT Y	GCC A	CAT H	GGA G	GCT A	ATG M	AAT N	ACT T	GGA G	TAC Y	AAT N	ATT I	GTG V	CTC L	TGG W	AGG R	GAG E	
Human (DNA/AA):	ATC I	AAT N	GGA G	CAC H	ATG M	ACT T	GTC V	ACT T	---	---	---	AAG K	ATG M	GCA A	GAA E	TAT Y	GAC D	CTA L	CAG Q	AAT N	GAT D	GTC V	TTC F	ATC I	ATC I	CCA P	GAT D	CAG Q	GAA E	ACA T	AAA K	AAT N	GAG E	TTC F	AGG R	AAT N	CTT L	AAG K	
Match Alignment:																																							
Bovine (DNA/AA):	ATC I	AAT N	GGC G	CAC H	ATG M	TCT S	ATC I	ACA T	AGA R	TGG W	CAC H	AAT N	ATG M	---	ACC T	TGA *	AGA R	ATG M	ATG M	TCT S	TCA S	TTG L	TCA S	CAA Q	ATC I	AAG K	AAA K	CAA Q	---	---	AAA K	ATG M	AGC S	ACA T	GGA G	ATC I	TTA L	---	

Bovine Reading Frame +2 — [FRAME-SHIFTED / MUTATED]

Human (DNA/AA):	TTA L	CTT L	GGT G	GTG V	CTA L	AAA K	AAT N	GTG V	---	---	ACA T	TTC F	ACT T	GAT D	GGA G	TGG W	---	AAT N	TCA S	TTT F	CAT H	TTT F	GAT D	GCT A	CAC H	GGG G	GAT D	TTA L	AAT N	ACT T	GGA G	TAT Y	GAT D	GTT V	GTG V	CTC L	TGG W	AAG K
Match Alignment:																																						
Bovine (DNA/AA):	--- -	--- -	--- -	GTA V	CTG L	GAG E	TGG W	GTT V	GCC A	ATT I	GCC A	TTC F	TCC S	GAT D	TCA S	TGG W	ATG M	GAG E	GGA G	GTT V	CAT H	TTC F	ATT I	TTT F	ATG M	CCC P	ATG M	GAG E	CTA L	TGA *	ATA I	CTG L	GAT D	ACA T	ATA I	TTG L	--- -	--- -
Human (DNA/AA):	GAG E	ATC I	AAT N	GGA G	CAC H	ATG M	ACT T	GTC V	ACT T	AAG K	ATG M	GCA A	GAA E	TAT Y	GAC D	CTA L	CAG Q	AAT N	GAT D	GTC V	TTC F	ATC I	ATC I	CCA P	GAT D	CAG Q	GAA E	--- -	--- -	--- -	ACA T	AAA K	AAT N	GAG E	TTC F	AGG R	AAT N	CTT L
Match Alignment:																																						
Bovine (DNA/AA):	--- -	TGC C	TCT S	GGA G	--- -	--- -	--- -	GGG G	AGA R	TCA S	ATG M	GCC A	--- -	ACA T	TGT C	CTA L	TCA S	CAA Q	GAT D	GGC G	ACA T	ATA I	TGA *	CCT P	--- -	GAA E	GAA E	TGA *	TGT C	CTT L	CAT H	TGT C	CAC H	AAA K	TCA S	AGA R	AAC N	AAA K

Human (DNA/AA):	AAG	---	---	---	---	---	---
	K	-	-	-	-	-	-
Match Alignment:							
Bovine (DNA/AA):	AAA	TGA	GCA	CAG	GAA	TCT	TAA
	K	*	A	Q	E	S	*

Bovine Reading Frame +3 — [CONSERVED & MATCHES MASS-SPEC BARCODE]

Human (DNA/AA):	TTA	CTT	GGT	GTG	CTA	AAA	AAT	GTG	ACA	TTC	ACT	GAT	GGA	TGG	---	---	---	AAT	TCA	TTT	CAT	TTT	GAT	GCT	CAC	GGG	GAT	TTA	AAT	ACT	GGA	TAT	GAT	GTT	GTG	CTC	TGG	AAG	
	L	L	G	V	L	K	N	V	T	F	T	D	G	W	-	-	-	N	S	F	H	F	D	A	H	G	D	L	N	T	G	Y	D	V	V	L	W	K	
Match Alignment:																																							
Bovine (DNA/AA):	TAC	TGG	AGT	GGG	TTG	CCA	TTG	CCT	TCT	CCG	ATT	CAT	GGA	TGG	AGG	GAG	TTC	ATT	TCA	TTT	TTA	TGC	CCA	TGG	AGC	TAT	GAA	TAC	TGG	ATA	CAA	TAT	---	TGT	GCT	CTG	---	---	
	Y	W	S	G	L	P	L	P	S	P	I	H	G	W	R	E	F	I	S	F	L	C	P	W	S	Y	E	Y	W	I	Q	Y	-	C	A	L	-	-	

Human (DNA/AA):	GAG	ATC	AAT	GGA	CAC	ATG	ACT	GTC	---	ACT	AAG	ATG	GCA	GAA	TAT	GAC	CTA	CAG	AAT	GAT	GTC	TTC	ATC	ATC	CCA	GAT	CAG	GAA	ACA	AAA	AAT	GAG	TTC	AGG	AAT	CTT	AAG
	E	I	N	G	H	M	T	V	-	T	K	M	A	E	Y	D	L	Q	N	D	V	F	I	I	P	D	Q	E	T	K	N	E	F	R	N	L	K
Match Alignment:																																					
Bovine (DNA/AA):	GAG	GGA	GAT	CAA	TGG	CCA	CAT	GTC	TAT	CAC	AAG	ATG	GCA	CAA	TAT	GAC	CTG	AAG	AAT	GAT	GTC	TTC	ATT	GTC	ACA	AAT	CAA	GAA	ACA	AAA	AAT	GAG	CAC	AGG	AAT	CTT	AAG
	E	G	D	Q	W	P	H	V	Y	H	K	M	A	Q	Y	D	L	K	N	D	V	F	I	V	T	N	Q	E	T	K	N	E	H	R	N	L	K

Concluding Diagnostic Framework:

Bovine Exon 4 translates natively into three prospective open reading frames. However, **Reading Frame +1** is verified as the sole functional configuration. It completely maintains evolutionary sequence alignment architecture with *Homo sapiens* and maps the full proteotypic mass spectrometry peptide "**NDVFIVTNQETK**" in perfect structural continuity. Reading Frames +2 and +3 yield premature stop codons and lack the base configurations needed to construct this evolutionary receptor domain, confirming strict translational frame conservation in the bull.